Designing a Model for Massive Digital Lifelong Learning

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ABSTRACT

A decade ago, massively open online courses (MOOCs) were heralded as the solution to universal global access to higher education. While they failed to reach this vision, primarily because of presentational/assimilative instruction (a PDF of the residential classroom), MOOCs provided foundational models and infrastructure for emergency remote learning in the pandemic. Reports of remote learning's death post-pandemic are greatly exaggerated, since the world is now irreversibly hybrid—and will stay that way because many people and organizations value the new opportunities this presents. From now on, all of us must have skills and attributes for adept, engaging performance both face-to-face and across distance. Moreover, online/hybrid interactions enable the multi-dimensional streams of data and natural language essential for personalizing instruction and motivating learning.

Further, as discussed in recent reports from Harvard, MIT, and Stanford, innovative approaches to digital learning were developed during the pandemic that are now improving campus-based learning [1]. Insights from these approaches offer the opportunity for student engagement at scale, taking advantage of strengths of online instruction such as collapsing time, bridging space, personalizing via rich patterns from micro-behaviors, using AI-based instructional assistants and learning partners, delivering content and experience across providers, and sustaining online learning communities that continue beyond formal instruction. Frameworks like Community of Inquiry (COI) can help organize these levers for engagement and learning [2].

Beyond transforming the process of instruction, shifting priorities to emphasize expanded learning outcomes is vital. The world faces a turbulent half-century in which adults must continuously upskill and reskill to succeed in work and life [3]. Many will have multiple careers, which means that cross-cutting skills like negotiation may have greater value than specific preparation for a job that could have a short half-life. Affective and social skills are gaining in importance, as the world is suffering from formal education that neglects capabilities such as empathy, xenophilia, social perspective taking, conflict resolution, and ethical decision making.

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Also, occupational roles and everyday life are increasingly permeated by artificial intelligence (AI) partners [4]. To attain Intelligence Augmentation (IA), in which a human being and AI partners each contribute their strengths, outcomes from lifelong learning must emphasize human judgment (practical wisdom) in complement to AI's reckoning (calculative prediction).

So profound a transformation of both process and outcomes requires redesign from the bottom-up, rather than adding onto our current educational structures. The lifelong learning business is very different than intermittent education centered on degrees, certificates, and performance on high-stakes psychometric tests. We have the advances we need to take on this Grand Challenge, moving beyond doing things better with formal education to doing better things via lifelong learning.

CCS CONCEPTS

• **Applied computing** → *Education*; • **Human-centered computing**; • **Social and professional topics** → *Computing/technology policy*

KEYWORDS

Lifelong learning, engagement at scale, community of inquiry, online/hybrid, artificial intelligence

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BIOGRAPHY



Chris Dede is a Senior Research Fellow at the Harvard Graduate School of Education and was for 22 years its Timothy E. Wirth Professor in Learning Technologies. His fields of scholarship include emerging technologies, policy, and leadership. From 2001-2004, he was Chair of the HGSE department of Teaching and Learning. In 2007, he was honored by Harvard University as an outstanding teacher, and in 2011 he was named a Fellow of the American Educational Research Association. In 2023 he was named a Fellow of the Online Learning Consortium.

Chris is a Co-Principal Investigator and Associate Director for Research of the NSF-funded National Artificial Intelligence Institute in Adult Learning and Online Education. In 2020 Chris co-founded

the Silver Lining for Learning initiative (https://silverliningforlearning.org). He is currently an Advisor to the Alliance for the Future of Digital Learning, sponsored by the Mohammed bin Rashid Global Initiative (MBRGI).

His most recent co-edited books include: *Teacher Learning in the Digital Age: Online Professional Development in STEM Education; Virtual, Augmented, and Mixed Realities in Education*; *Learning engineering for online education: Theoretical contexts and design-based examples;* and *The 60-Year Curriculum: New Models for Lifelong Learning in the Digital Economy.*

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